Pathology prediction from the EEG: status and diagnosis groups

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This notebook reads the classification side of the multi-task brain-age models: how well the EEG predicts the **normal/abnormal status** and **13 NEDC diagnosis groups** of a TUH recording. For every target it reports the threshold-free ranking (ROC-AUC, PR-AUC), the **best F1** over all thresholds with its bootstrap interval, the **errors** (false positives and negatives) at that threshold, and the **ROC and precision-recall curves**.

It compares four runs on **one identical split** (Exp4mtbal's recorded `data_split.json`: 27,616 train, 3,422 validation and 274 TUAB-test recordings):

- two runs that also predict age (`Exp4mt` , `Exp4mtbal` , Table 2.10 of `brain_age_research_summary.ipynb`);
- two **classification-only** runs without the age prediction: `Exp4cls` (EEG only) and `Exp4clsage` (the recording's known age as an extra encoder input).

Per-recording probabilities come from `scripts/eval_multitask_classification.py` (`docs/data/multitask_class_predictions.csv.gz`); every metric is computed here from them by `labram/eval/binary_metrics.py` .

```
runs scored: ['Exp4mt', 'Exp4mtbal', 'Exp4cls', 'Exp4clsage']
```

1. Runs and labels

Table 1.1. The runs

All four runs share everything except the columns below: `Exp4noearEEG` 's setup (19 channels without A1/A2/T1/T2, EEG-only input, LaBraM-base start, 8 epochs with 2 warm-up, 5 training minutes per recording per epoch, mixup, early stopping) and the multi-task head (2 shared layers 256-256, an age branch 128-64-1 and a classification branch 128-64-14). The classification loss is $0.5\,\mathrm{BCE}_{\mathrm{status}} + 0.5\,L_{\mathrm{path}}$ with L_{path} the plain (`known`) or class-balanced (`present`) mean BCE over the diagnosis groups (Eqs. (2.15)–(2.16) of the research summary). `age weight` multiplies the age (Huber) loss (`loss.age_weight`); at 0 the age branch gets no gradient and the model is a pure classifier. `age input` `tokens` (`model.age_input=tokens`) feeds the recording's known age (z-scored with the training mean 52.35 and SD 17.60 years) through a `Linear(1 → 200)` + LayerNorm as one extra token into the transformer, next to the EEG patch tokens. `selected on` is the validation metric that picks the reported epoch.

run	age weight	pathology loss	age input	selected on	where
`Exp4mt`	1	known	–	age MAE	SageMaker
`Exp4mtbal`	1	present (class-balanced)	–	age MAE	local
`Exp4cls`	**0**	present	–	pathology macro ROC-AUC	local
`Exp4clsage`	**0**	present	**tokens**	pathology macro ROC-AUC	SageMaker

Table 1.2. Labels per target

Per target: the recordings whose label is known, the positives among them and their prevalence π (Eq. (2.6)), on validation and on TUAB test. A label is unknown when the recording has no status, or for a group when its session is not in the NEDC clinical sheet; unknown labels are never scored. TUAB test holds only 11 diagnosis-group positives in total, so the groups are judged on validation and test is used for the status.

target	val known	val positives	val prevalence %	test known	test positives
status_abnormal	3064	1676	54.70	274	125
clinical_seizure	2991	185	6.19	268	0
electrographic_seizure	2991	11	0.37	268	0
epilepsy	2991	46	1.54	268	0
stroke	2991	93	3.11	268	1
tbi	2991	50	1.67	268	0
substance	2991	103	3.44	268	10
bipolar	2991	31	1.04	268	7
anxiety	2991	10	0.33	268	0
depression	2991	8	0.27	268	0
schizophrenia	2991	9	0.30	268	0
dementia	2991	10	0.33	268	1
ms	2991	7	0.23	268	0
other	2991	12	0.40	268	0

2. Metrics

For one target, recording i of a split has the label $y_i \in \{0, 1\}$ (known labels only) and the score p_i, the mean of its windows' sigmoid probabilities (as in training's per-recording metrics). Predicting positive when $p_i \geq t$ gives, at threshold t, the true and false positives and negatives $\mathrm{TP}(t), \mathrm{FP}(t), \mathrm{FN}(t), \mathrm{TN}(t)$ (counts of recordings, `binary_metrics.py::threshold_metrics`) and

$$\mathrm{prec}(t) = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad \mathrm{rec}(t) = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \qquad \mathrm{spec}(t) = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \tag{2.1}$$

$$F_1(t) = \frac{2\,\mathrm{prec}(t)\,\mathrm{rec}(t)}{\mathrm{prec}(t) + \mathrm{rec}(t)} \tag{2.2}$$

(0 when no recording is flagged). The **errors** at t are the error rate and the balanced error

$$\mathrm{err}(t) = \frac{\mathrm{FP} + \mathrm{FN}}{n}, \qquad \mathrm{berr}(t) = 1 - \tfrac{1}{2}\big(\mathrm{rec}(t) + \mathrm{spec}(t)\big) \tag{2.3}$$

with n the known recordings. The **best F1** is the maximum over every distinct score as threshold (`best_f1_threshold`; ties go to the higher threshold):

$$t^\star = \arg\max_{t \in \{p_i\}} F_1(t), \qquad F_1^\star = F_1(t^\star) \tag{2.4}$$

$F_1^\star$ on a split picks $t^\star$ on that split's own labels, so it is optimistic. The honest operating point on test carries the **validation** threshold over:

$$F_1^{\text{val}\to\text{test}} = F_1^{\text{test}}\left(t^\star_{\text{val}}\right) \tag{2.5}$$

The threshold-free metrics are the ROC-AUC (`sklearn` `roc_auc_score` , the area under $\text{rec}(t)$ against $1 - \text{spec}(t)$) and the PR-AUC as average precision (`average_precision_score`):

$$\text{AP} = \sum_k \left(\text{rec}_k - \text{rec}_{k-1}\right) \text{prec}_k, \qquad \mathbb{E}[\text{AP}_{\text{random}}] = \pi = \frac{1}{n}\sum_i y_i \tag{2.6}$$

over the thresholds k in decreasing order. A random score has AP $\approx \pi$ and ROC-AUC 0.5; flagging every recording gives $F_1 = 2\pi/(1+\pi)$, the floor marked in Figure 3.1. With 7–50 positives per rare group, every metric carries a **95 % stratified bootstrap interval** (`bootstrap_ci` , $B = 1000$, positives and negatives resampled separately, $t^\star$ re-chosen in each resample):

$$\left[\, q_{0.025}\{m^{(b)}\}_{b=1}^{B},\ q_{0.975}\{m^{(b)}\}_{b=1}^{B} \,\right] \tag{2.7}$$

with $m^{(b)}$ the metric on resample b and q the empirical quantile. The macro value of a metric is its mean over the 13 groups (status excluded):

$$m_{\text{macro}} = \frac{1}{13}\sum_{k=1}^{13} m_k \tag{2.8}$$

3. Results

Table 3.1. Ranking quality on validation: ROC-AUC and PR-AUC

Per target and run, on the 3,422 validation recordings (known labels only, Table 1.2): ROC-AUC and PR-AUC (Eq. (2.6)) with their 95 % bootstrap intervals (Eq. (2.7)) as `value [low, high]` ; `prevalence` is the PR-AUC of a random score. The last row is the macro mean over the 13 groups (Eq. (2.8)).

target	preva-lence	ROC-AUC Exp4mt	ROC-AUC Exp4mtbal	ROC-AUC Exp4cls	ROC-AUC Exp4clsage	PR-AUC Exp4mt	PR-AUC Exp4mtbal	PR-AUC Exp4cls	PR-AUC Exp4clsage
status_abnormal	0.547	0.822 [0.806, 0.836]	0.820 [0.804, 0.836]	0.813 [0.797, 0.829]	0.827 [0.812, 0.841]	0.821 [0.802, 0.840]	0.819 [0.799, 0.838]	0.807 [0.786, 0.828]	0.825 [0.806, 0.844]
clinical_seizure	0.062	0.778 [0.750, 0.805]	0.789 [0.760, 0.819]	0.783 [0.754, 0.809]	0.785 [0.755, 0.815]	0.180 [0.148, 0.227]	0.204 [0.166, 0.256]	0.183 [0.152, 0.233]	0.183 [0.155, 0.229]
electrographic_seizure	0.004	0.963 [0.933, 0.985]	0.962 [0.947, 0.975]	0.964 [0.943, 0.983]	0.983 [0.969, 0.993]	0.083 [0.047, 0.199]	0.051 [0.039, 0.083]	0.106 [0.041, 0.330]	0.231 [0.103, 0.454]
epilepsy	0.015	0.740 [0.692, 0.786]	0.680 [0.620, 0.739]	0.698 [0.656, 0.742]	0.709 [0.639, 0.776]	0.030 [0.025, 0.041]	0.024 [0.020, 0.032]	0.024 [0.021, 0.030]	0.034 [0.026, 0.049]
stroke	0.031	0.756 [0.706, 0.800]	0.769 [0.720, 0.813]	0.795 [0.751, 0.839]	0.732 [0.683, 0.777]	0.083 [0.067, 0.116]	0.105 [0.077, 0.164]	0.143 [0.107, 0.210]	0.072 [0.058, 0.100]
tbi	0.017	0.764 [0.713, 0.813]	0.742 [0.681, 0.799]	0.761 [0.697, 0.823]	0.694 [0.633, 0.757]	0.046 [0.033, 0.086]	0.052 [0.034, 0.111]	0.117 [0.066, 0.230]	0.033 [0.026, 0.050]
substance	0.034	0.618 [0.557, 0.674]	0.611 [0.552, 0.666]	0.629 [0.571, 0.684]	0.588 [0.532, 0.644]	0.057 [0.045, 0.078]	0.058 [0.044, 0.088]	0.053 [0.044, 0.069]	0.048 [0.040, 0.065]
bipolar	0.010	0.556 [0.471, 0.641]	0.532 [0.448, 0.624]	0.654 [0.560, 0.734]	0.571 [0.479, 0.668]	0.012 [0.010, 0.016]	0.011 [0.009, 0.016]	0.020 [0.013, 0.038]	0.015 [0.011, 0.030]
anxiety	0.003	0.740 [0.554, 0.895]	0.770 [0.597, 0.914]	0.734 [0.561, 0.872]	0.744 [0.573, 0.889]	0.031 [0.007, 0.145]	0.035 [0.008, 0.165]	0.012 [0.006, 0.040]	0.019 [0.007, 0.077]
depression	0.003	0.633 [0.372, 0.865]	0.641 [0.421, 0.851]	0.558 [0.304, 0.792]	0.580 [0.378, 0.789]	0.011 [0.004, 0.040]	0.009 [0.003, 0.042]	0.005 [0.003, 0.013]	0.006 [0.003, 0.024]
schizophrenia	0.003	0.551 [0.487, 0.622]	0.703 [0.628, 0.774]	0.735 [0.618, 0.835]	0.773 [0.698, 0.837]	0.003 [0.003, 0.005]	0.005 [0.005, 0.008]	0.006 [0.005, 0.012]	0.007 [0.006, 0.012]
dementia	0.003	0.848 [0.779, 0.910]	0.890 [0.851, 0.928]	0.865 [0.813, 0.915]	0.863 [0.778, 0.929]	0.015 [0.010, 0.045]	0.025 [0.013, 0.105]	0.014 [0.010, 0.028]	0.017 [0.010, 0.036]
ms	0.002	0.522 [0.364, 0.682]	0.642 [0.424, 0.806]	0.538 [0.451, 0.636]	0.635 [0.510, 0.752]	0.003 [0.002, 0.007]	0.004 [0.003, 0.009]	0.003 [0.003, 0.004]	0.004 [0.003, 0.007]
other	0.004	0.623 [0.493, 0.735]	0.652 [0.539, 0.761]	0.585 [0.480, 0.694]	0.580 [0.479, 0.688]	0.006 [0.005, 0.009]	0.006 [0.005, 0.010]	0.005 [0.004, 0.011]	0.005 [0.004, 0.010]
macro (13 groups)	–	0.699	0.722	0.715	0.711	0.043	0.045	0.053	0.052

Table 3.2. Best F1 and the errors at its threshold, validation

Per target and run: the best F1 $F_1^\star$ (Eq. (2.4)) with its 95 % interval (Eq. (2.7)), the threshold $t^\star$ (a probability), precision and recall at it (Eq. (2.1)), the errors as recording counts — false positives FP and false negatives FN next to the true positives TP — and the error rate and balanced error (Eq. (2.3)) in percent. The last row per run is the macro mean of $F_1^\star$ (Eq. (2.8)).

run	target	best F1	t*	precision	recall	TP	FP	FN	error %	balanced error %
	status_abnormal	0.799 [0.789, 0.812]	0.43	0.75	0.86	1436	481	240	23.53	24.49
	clinical_seizure	0.247 [0.225, 0.288]	0.124	0.17	0.46	85	418	100	17.32	34.48
	electrographic_seizure	0.182 [0.114, 0.342]	0.098	0.11	0.46	5	39	6	1.50	27.93
	epilepsy	0.067 [0.058, 0.100]	0.0267	0.04	0.67	31	843	15	28.69	30.62
	stroke	0.162 [0.142, 0.206]	0.0528	0.09	0.59	55	530	38	18.99	29.57
	tbi	0.107 [0.071, 0.187]	0.0225	0.06	0.34	17	252	33	9.53	37.28
	substance	0.132 [0.096, 0.187]	0.0778	0.11	0.17	18	151	85	7.89	43.88
Exp4mt	bipolar	0.028 [0.023, 0.042]	0.0109	0.01	0.74	23	1615	8	54.26	40.18
	anxiety	0.148 [0.018, 0.345]	0.0253	0.12	0.20	2	15	8	0.77	40.25
	depression	0.049 [0.014, 0.121]	0.0137	0.03	0.25	2	71	6	2.57	38.69
	schizophrenia	0.010 [0.010, 0.012]	0.00145	0.01	1.00	9	1800	0	60.18	30.18
	dementia	0.050 [0.025, 0.158]	0.0226	0.03	0.10	1	29	9	1.27	45.49
	ms	0.008 [0.006, 0.024]	0.0036	0.00	0.14	1	242	6	8.29	46.91
	other	0.016 [0.012, 0.024]	0.000802	0.01	0.83	10	1265	2	42.36	29.57
	macro (13 groups)	0.093	–	0.06	0.46	–	–	–	19.51	36.54
	status_abnormal	0.795 [0.785, 0.808]	0.416	0.74	0.85	1432	493	244	24.05	25.04
	clinical_seizure	0.270 [0.236, 0.319]	0.159	0.20	0.43	79	321	106	14.28	34.37
	electrographic_seizure	0.125 [0.086, 0.205]	0.0766	0.07	0.64	7	94	4	3.28	19.76
	epilepsy	0.062 [0.048, 0.085]	0.0409	0.03	0.43	20	584	26	20.39	38.18
	stroke	0.163 [0.145, 0.232]	0.0806	0.10	0.37	34	291	59	11.70	36.74
	tbi	0.095 [0.072, 0.206]	0.0439	0.08	0.12	6	70	44	3.81	45.19
	substance	0.127 [0.091, 0.196]	0.0656	0.09	0.21	22	221	81	10.10	43.15
Exp4mtbal	bipolar	0.025 [0.023, 0.044]	0.018	0.01	0.45	14	1070	17	36.34	45.49
	anxiety	0.143 [0.024, 0.353]	0.0678	0.11	0.20	2	16	8	0.80	40.27
	depression	0.043 [0.009, 0.138]	0.043	0.03	0.12	1	38	7	1.50	44.39
	schizophrenia	0.015 [0.012, 0.020]	0.0108	0.01	0.78	7	902	2	30.22	26.23
	dementia	0.100 [0.031, 0.300]	0.102	0.10	0.10	1	9	9	0.60	45.15
	ms	0.012 [0.007, 0.025]	0.0122	0.01	0.57	4	644	3	21.63	32.22
	other	0.016 [0.012, 0.026]	0.0126	0.01	0.50	6	710	6	23.94	36.92
	macro (13 groups)	0.092	–	0.07	0.38	–	–	–	13.74	37.54
	status_abnormal	0.790 [0.778, 0.804]	0.474	0.76	0.82	1377	434	299	23.92	24.55
	clinical_seizure	0.241 [0.224, 0.297]	0.227	0.23	0.25	46	150	139	9.66	40.24
	electrographic_seizure	0.154 [0.096, 0.429]	0.279	0.50	0.09	1	1	10	0.37	45.47
	epilepsy	0.058 [0.051, 0.072]	0.0189	0.03	0.76	35	1129	11	38.11	31.12
	stroke	0.256 [0.192, 0.351]	0.159	0.28	0.24	22	57	71	4.28	39.16
	tbi	0.198 [0.142, 0.322]	0.0778	0.26	0.16	8	23	42	2.17	42.39
	substance	0.113 [0.095, 0.151]	0.0482	0.07	0.42	43	614	60	22.53	39.76
Exp4cls	bipolar	0.072 [0.032, 0.145]	0.031	0.05	0.13	4	76	27	3.44	44.83
	anxiety	0.056 [0.018, 0.133]	0.045	0.03	0.20	2	59	8	2.24	40.99
	depression	0.015 [0.007, 0.040]	0.0169	0.01	0.50	4	518	4	17.45	33.68
	schizophrenia	0.016 [0.011, 0.031]	0.0121	0.01	0.56	5	601	4	20.23	32.30
	dementia	0.037 [0.025, 0.077]	0.0272	0.02	0.30	3	147	7	5.15	37.47
	ms	0.008 [0.007, 0.010]	0.00752	0.00	1.00	7	1815	0	60.68	30.41
	other	0.013 [0.011, 0.040]	0.00785	0.01	0.75	9	1383	3	46.34	35.71
	macro (13 groups)	0.095	–	0.12	0.41	–	–	–	17.90	37.96
	status_abnormal	0.798 [0.787, 0.811]	0.412	0.74	0.86	1439	493	237	23.82	24.83
	clinical_seizure	0.270 [0.237, 0.321]	0.136	0.20	0.42	77	308	108	13.91	34.68
	electrographic_seizure	0.286 [0.203, 0.516]	0.105	0.21	0.46	5	19	6	0.84	27.59
	epilepsy	0.094 [0.070, 0.131]	0.0572	0.05	0.35	16	280	30	10.36	37.36
	stroke	0.135 [0.121, 0.192]	0.0484	0.08	0.55	51	610	42	21.80	33.10
	tbi	0.093 [0.067, 0.142]	0.0238	0.05	0.36	18	319	32	11.73	37.42
	substance	0.109 [0.084, 0.157]	0.0601	0.07	0.23	24	312	79	13.07	43.75
Exp4clsage	bipolar	0.056 [0.025, 0.113]	0.0317	0.03	0.16	5	141	26	5.58	44.32
	anxiety	0.103 [0.017, 0.250]	0.0632	0.07	0.20	2	27	8	1.17	40.45
	depression	0.028 [0.007, 0.088]	0.0409	0.02	0.12	1	63	7	2.34	44.81
	schizophrenia	0.017 [0.014, 0.030]	0.016	0.01	0.67	6	677	3	22.73	28.02
	dementia	0.054 [0.028, 0.095]	0.0205	0.03	0.50	5	170	5	5.85	27.85
	ms	0.009 [0.007, 0.016]	0.0113	0.00	0.86	6	1381	1	46.20	30.28
	other	0.012 [0.011, 0.031]	0.007	0.01	0.83	10	1709	2	57.20	37.02
	macro (13 groups)	0.097	–	0.06	0.44	–	–	–	16.37	35.90

Table 3.3. TUAB test: best F1 and the validation threshold carried over

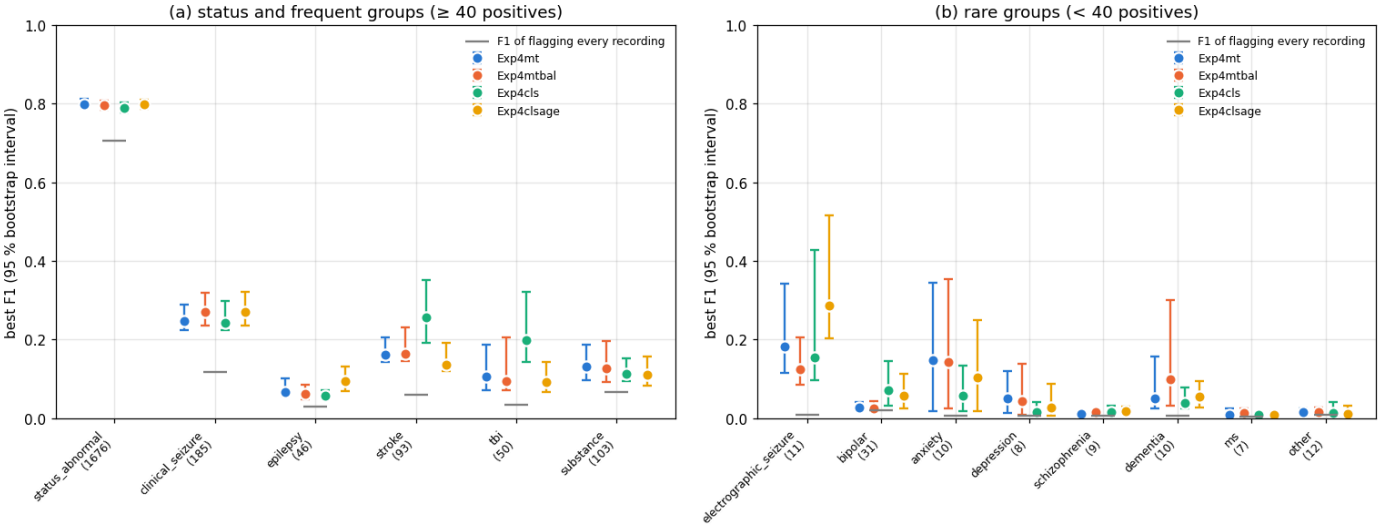
The 274 TUAB-test recordings, for the targets with test positives: ROC-AUC, the best F1 on test $F_1^{\star}$ (Eq. (2.4), optimistic: its threshold is chosen on test), and the honest $F_1^{\mathrm{val}\to\mathrm{test}}$ (Eq. (2.5))

with its errors (FP, FN) and balanced error (Eq. (2.3)). Only the status has enough test positives (125) to read; the groups have 1–10.

target	run	positives	ROC-AUC	best F1 (test t*)	F1 at val t*	val t*	TP	FP	FN	balanced error % at val t*
status_abnormal	Exp4mt	125	0.92	0.85	0.82	0.43	111	36	14	17.68
	Exp4mtbal	125	0.92	0.84	0.82	0.416	111	34	14	17.01
	Exp4cls	125	0.91	0.83	0.79	0.474	100	28	25	19.40
	Exp4clsage	125	0.93	0.85	0.83	0.412	111	31	14	16.00
stroke	Exp4mt	1	0.73	0.03	0.00	0.0528	0	36	1	56.74
	Exp4mtbal	1	0.76	0.03	0.00	0.0806	0	19	1	53.56
	Exp4cls	1	0.96	0.15	0.00	0.159	0	0	1	50.00
	Exp4clsage	1	0.98	0.25	0.06	0.0484	1	31	0	5.80
substance	Exp4mt	10	0.51	0.14	0.12	0.0778	1	5	9	45.97
	Exp4mtbal	10	0.53	0.12	0.00	0.0656	0	16	10	53.10
	Exp4cls	10	0.54	0.14	0.14	0.0482	4	44	6	38.53
	Exp4clsage	10	0.43	0.13	0.12	0.0601	1	6	9	46.16
bipolar	Exp4mt	7	0.72	0.13	0.07	0.0109	6	161	1	37.99
	Exp4mtbal	7	0.71	0.20	0.08	0.018	5	119	2	37.08
	Exp4cls	7	0.62	0.10	0.00	0.031	0	8	7	51.53
	Exp4clsage	7	0.74	0.11	0.10	0.0317	2	31	5	41.65
dementia	Exp4mt	1	0.90	0.07	0.00	0.0226	0	2	1	50.38
	Exp4mtbal	1	0.93	0.09	0.00	0.102	0	2	1	50.38
	Exp4cls	1	0.79	0.04	0.00	0.0272	0	6	1	51.12
	Exp4clsage	1	0.69	0.02	0.00	0.0205	0	2	1	50.38

Figure 3.1. Best F1 per target with its bootstrap interval, validation

$F_1^\star$ (Eq. (2.4)) per target and run with its 95 % interval (Eq. (2.7)) on the 3,422 validation recordings; the number under each target is its positives. (a) Status and the groups with at least 40 positives; (b) the rare groups. The grey bar is the F1 of flagging every recording, $2\pi/(1+\pi)$ (Eq. (2.6)): a model must clear it to be of any use as a detector.



Figures 3.2–3.15. ROC and precision-recall curves per target

One figure per target (status first, then the 13 groups), on validation (3,422 recordings, known labels only); each run is one line in its fixed colour, the dot is its best-F1 operating point $t^\star$ (Eq. (2.4)). (a) ROC curve: recall against the false-positive rate $1 - \mathrm{spec}$ (Eq. (2.1)), chance on the

diagonal; legend: ROC-AUC. (b) Precision-recall curve; chance is the prevalence π (Eq. (2.6)), the grey contours are constant F1 (Eq. (2.2)); legend: average precision and $F_1^\star$. The status figure (3.2) is followed by its TUAB-test version (3.2(test), 274 recordings).

Figure 3.2. status_abnormal: ROC and precision-recall curves, validation

1676 positives of 3064 recordings (prevalence 0.547).

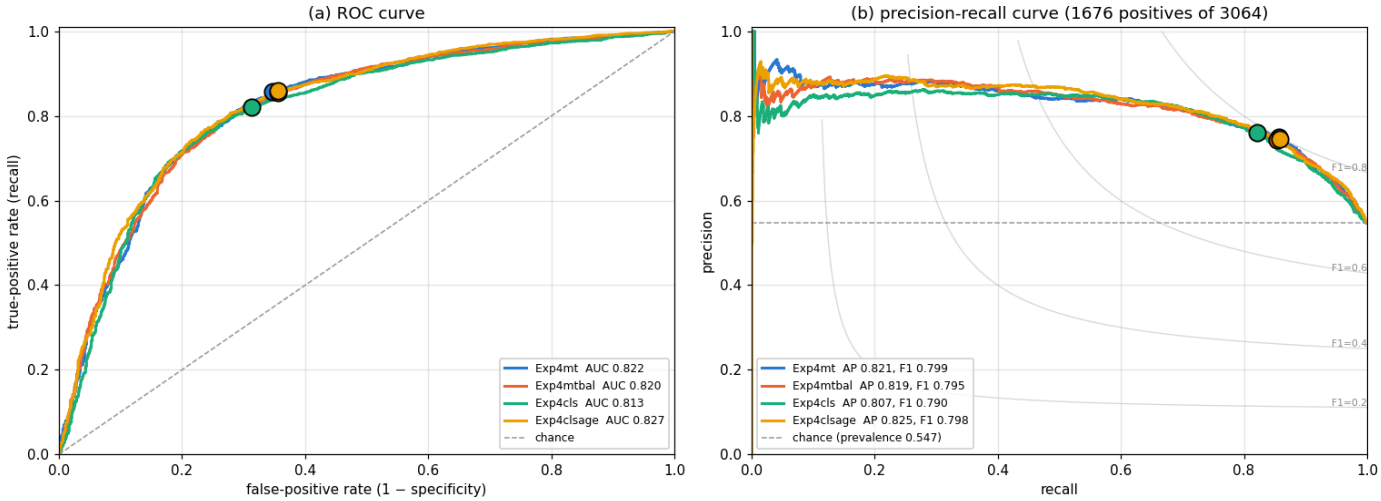


Figure 3.2(test). status_abnormal: ROC and precision-recall curves, TUAB test

274 TUAB-test recordings, 125 abnormal.

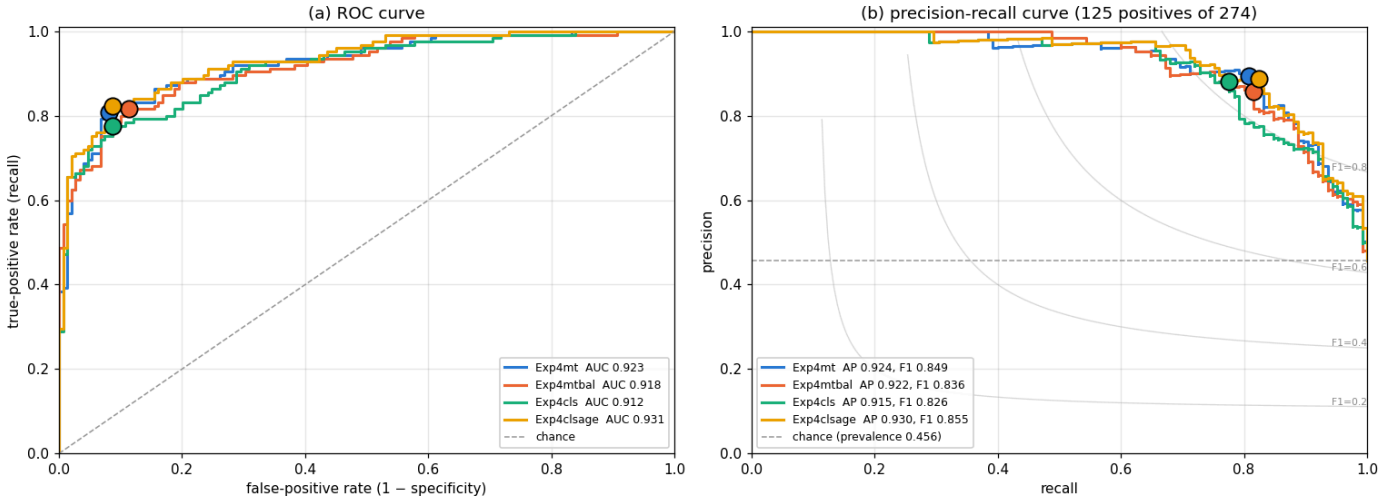


Figure 3.3. clinical_seizure: ROC and precision-recall curves, validation

185 positives of 2991 recordings (prevalence 0.062).

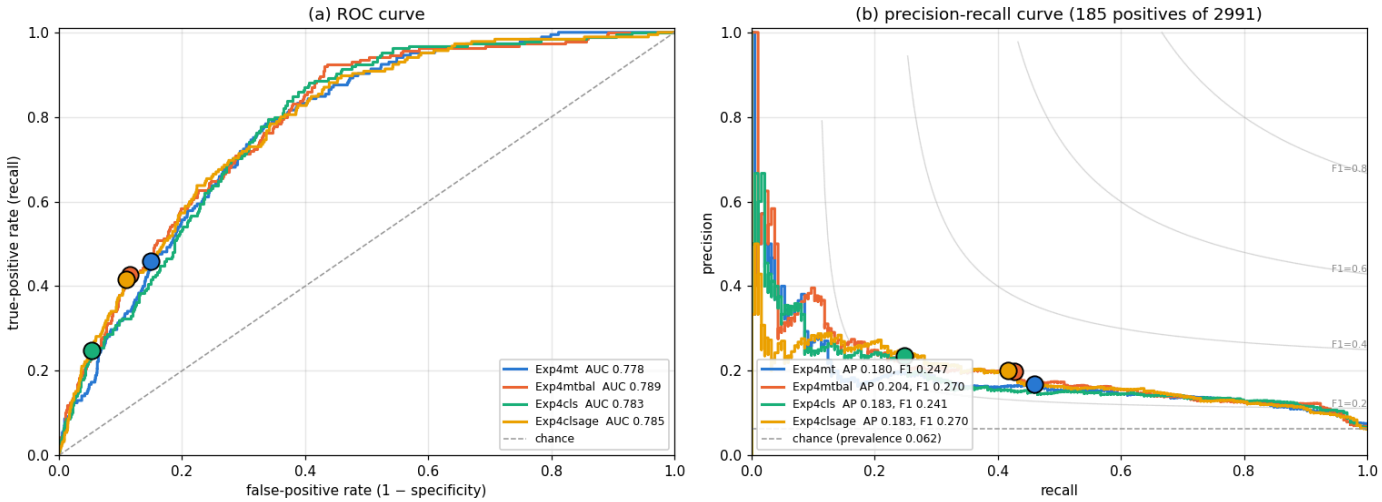


Figure 3.4. electrographic_seizure: ROC and precision-recall curves, validation

11 positives of 2991 recordings (prevalence 0.004).

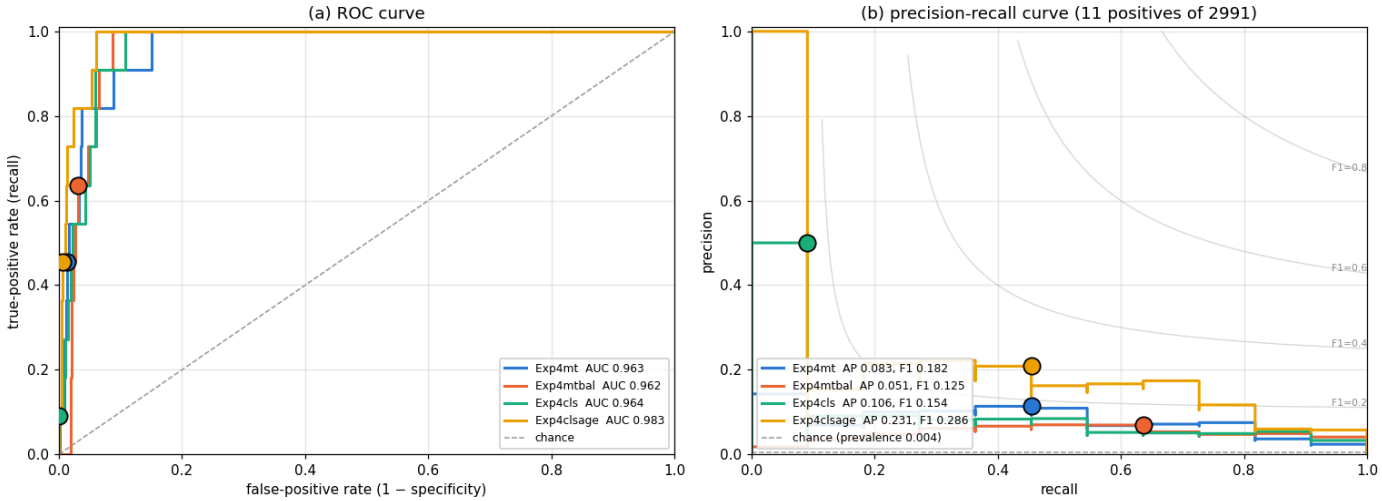


Figure 3.5. epilepsy: ROC and precision-recall curves, validation

46 positives of 2991 recordings (prevalence 0.015).

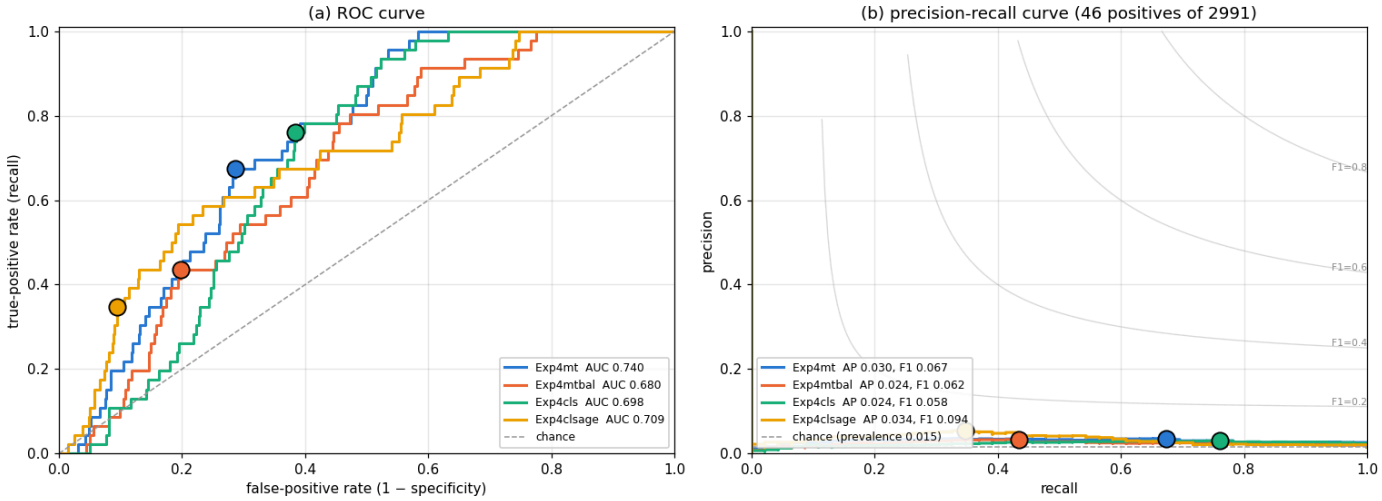


Figure 3.6. stroke: ROC and precision-recall curves, validation

93 positives of 2991 recordings (prevalence 0.031).

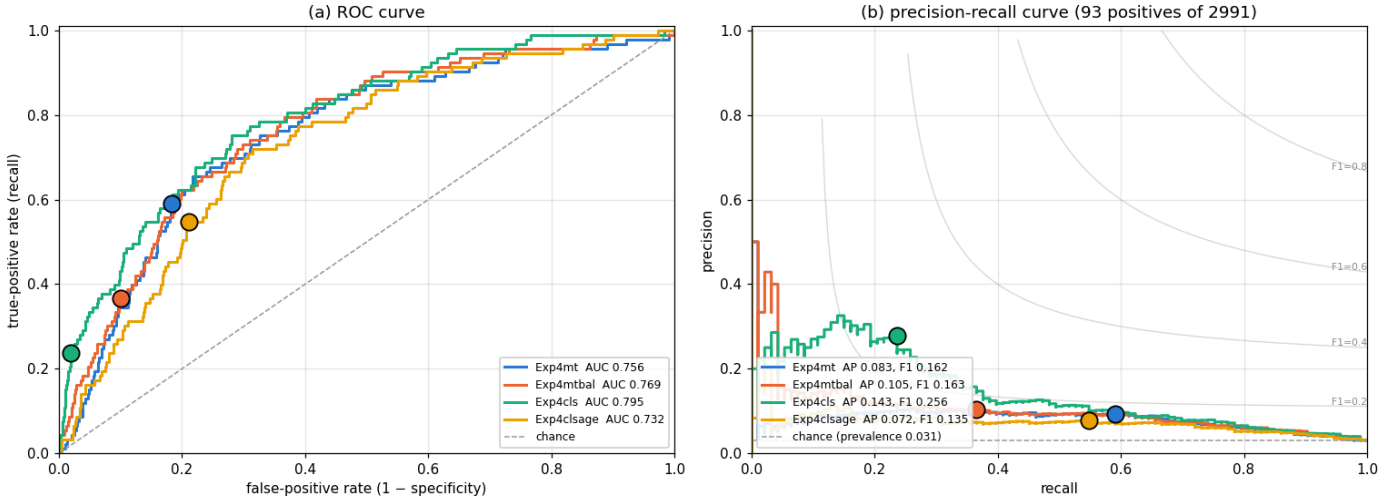


Figure 3.7. tbi: ROC and precision-recall curves, validation

50 positives of 2991 recordings (prevalence 0.017).

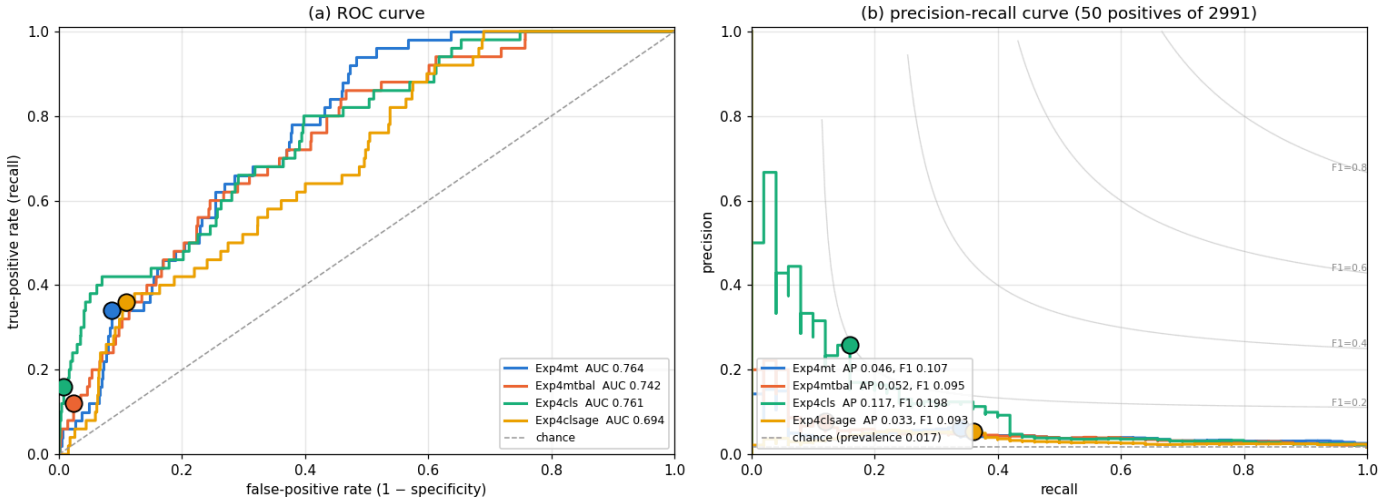


Figure 3.8. substance: ROC and precision-recall curves, validation

103 positives of 2991 recordings (prevalence 0.034).

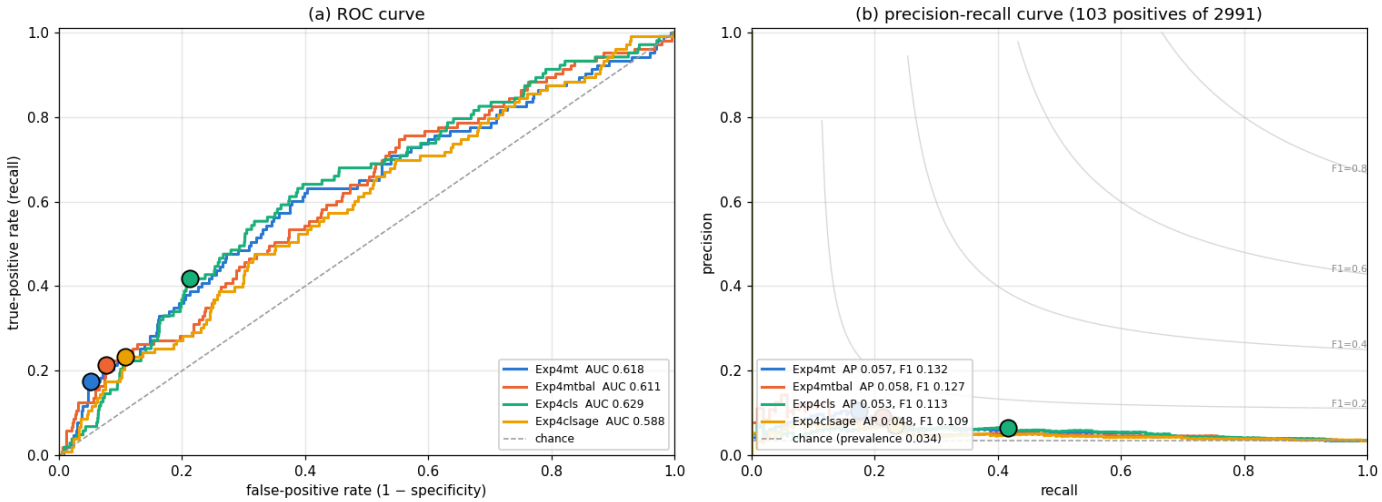


Figure 3.9. bipolar: ROC and precision-recall curves, validation

31 positives of 2991 recordings (prevalence 0.010).

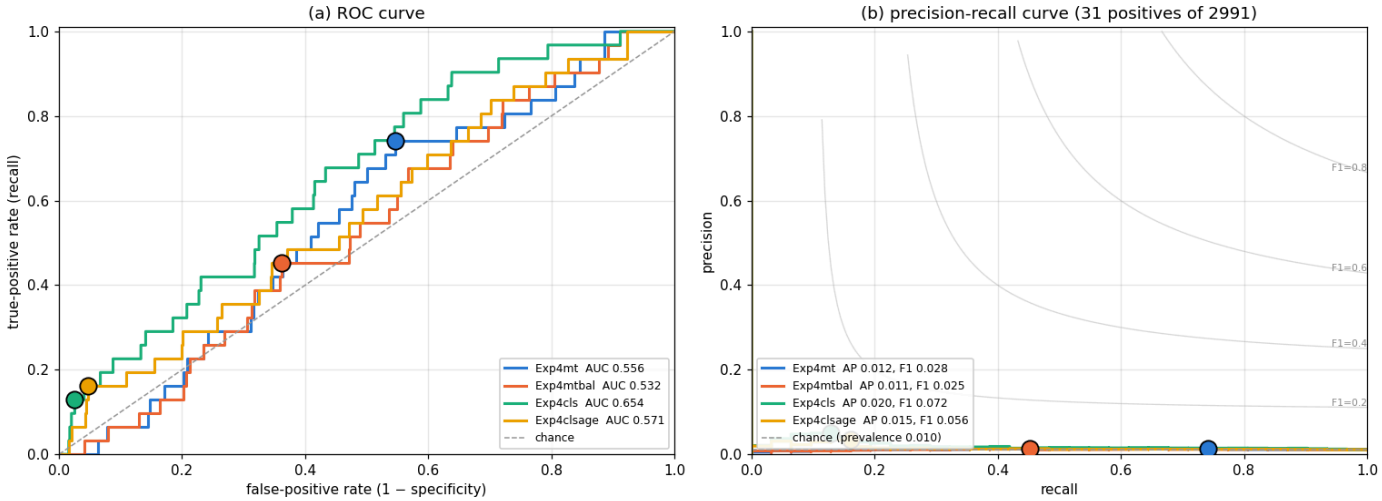


Figure 3.10. anxiety: ROC and precision-recall curves, validation

10 positives of 2991 recordings (prevalence 0.003).

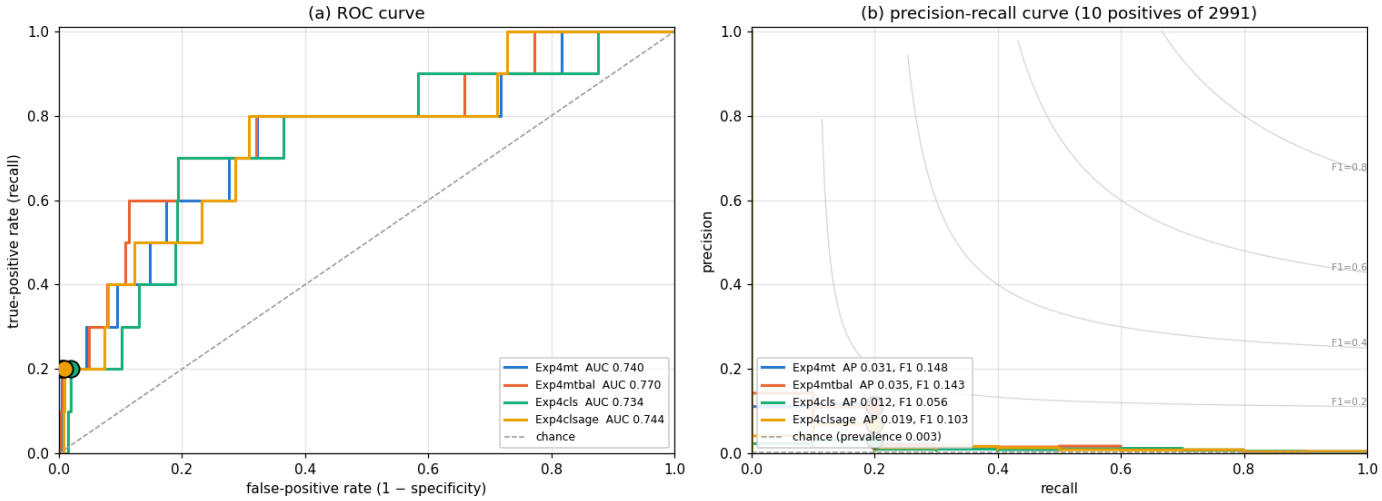


Figure 3.11. depression: ROC and precision-recall curves, validation

8 positives of 2991 recordings (prevalence 0.003).

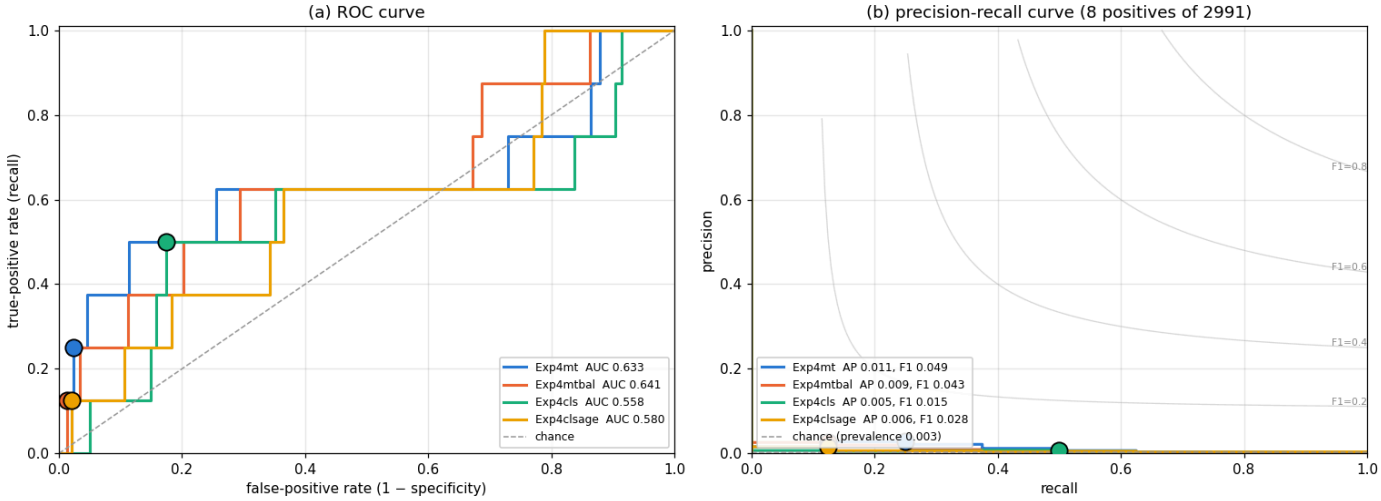


Figure 3.12. schizophrenia: ROC and precision-recall curves, validation

9 positives of 2991 recordings (prevalence 0.003).

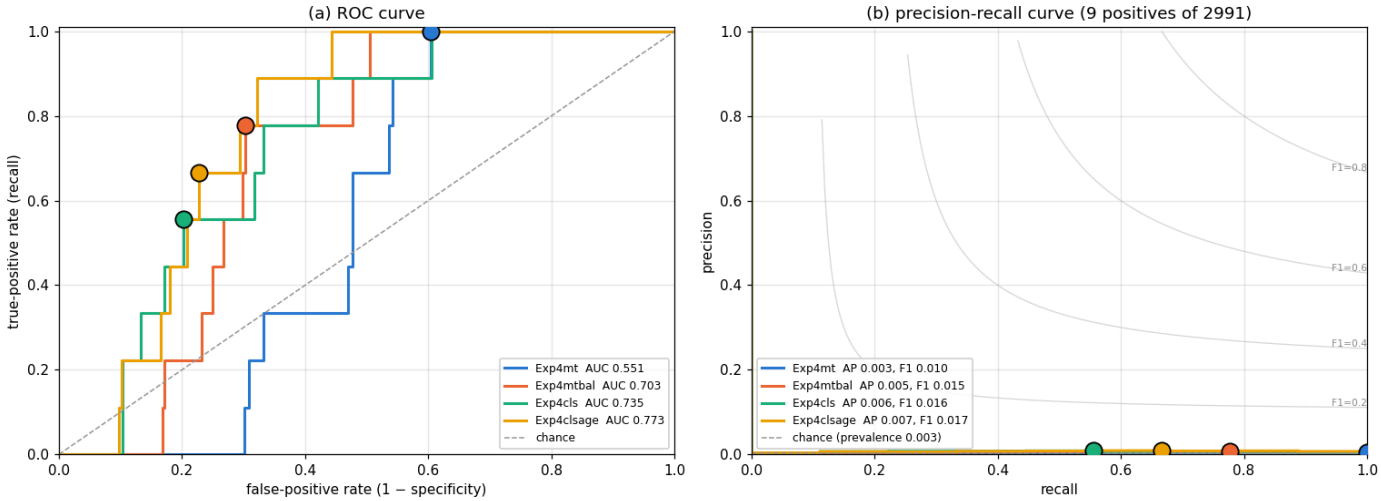


Figure 3.13. dementia: ROC and precision-recall curves, validation

10 positives of 2991 recordings (prevalence 0.003).

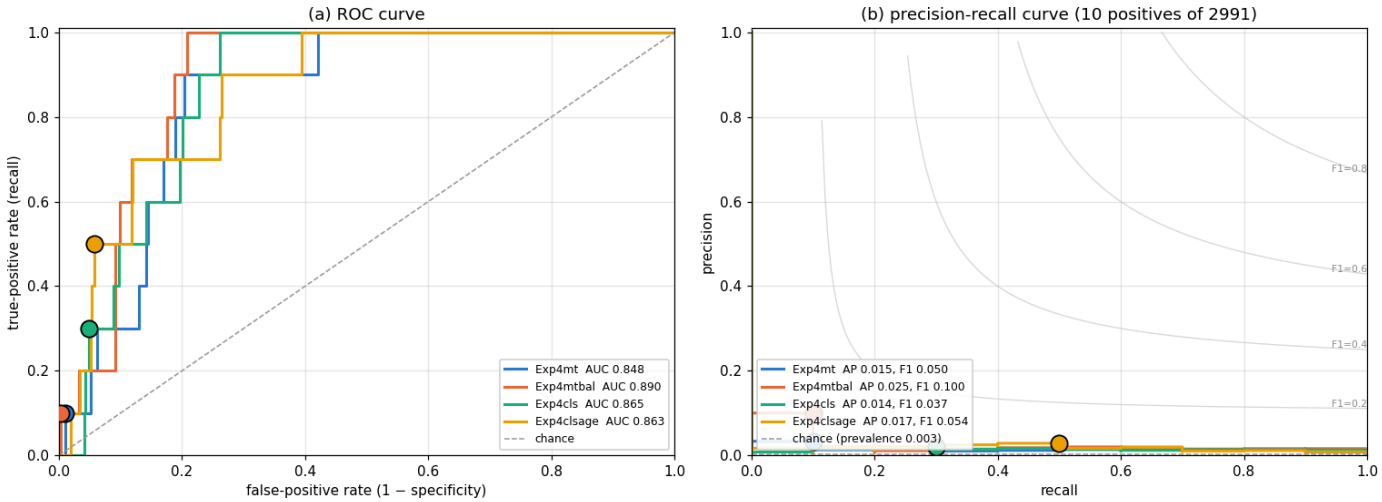


Figure 3.14. ms: ROC and precision-recall curves, validation

7 positives of 2991 recordings (prevalence 0.002).

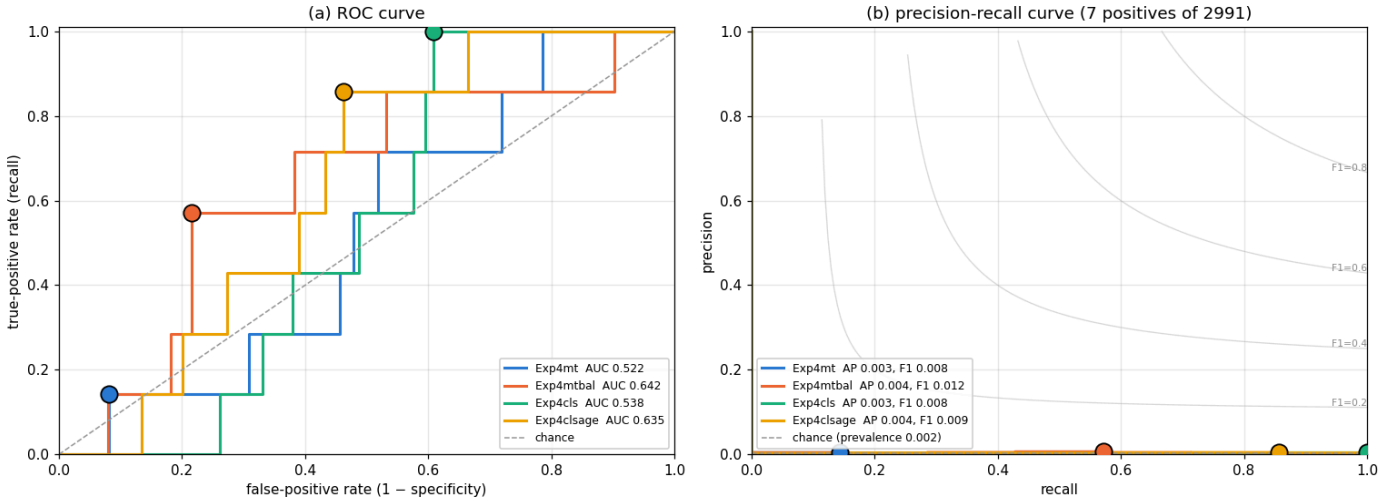
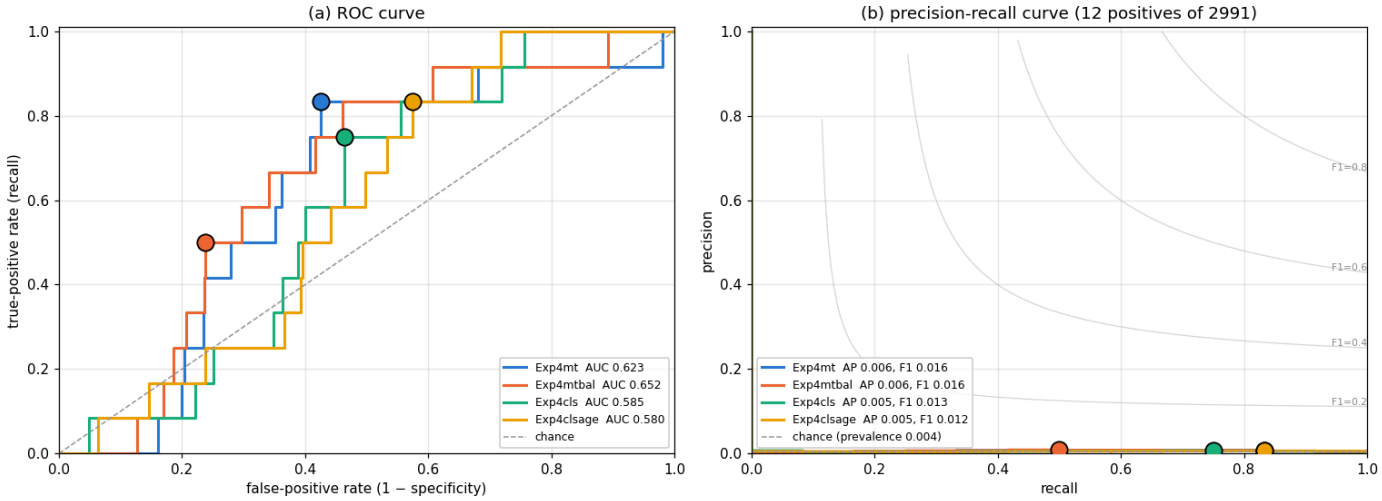


Figure 3.15. other: ROC and precision-recall curves, validation

12 positives of 2991 recordings (prevalence 0.004).



4. Reading the results

Table 4.1. Classification-only runs against the age-predicting run with the same loss

Each later run minus `Exp4mtbal` (same class-balanced loss, same split), on validation: ROC-AUC, PR-AUC and $F_1^\star$ of the status, and their macro means over the 13 groups (Eq. (2.8)). Positive = better than `Exp4mtbal`. For a single rare group a difference below about 0.1 ROC-AUC is inside its bootstrap interval (Table 3.1); the macro values average that noise down.

	status ROC-AUC	macro ROC-AUC	status PR-AUC	macro PR-AUC	status best F1	macro best F1
Exp4mt − Exp4mtbal	+0.001	-0.022	+0.002	-0.002	+0.004	+0.001
Exp4cls − Exp4mtbal	-0.007	-0.007	-0.012	+0.008	-0.006	+0.003
Exp4clsage − Exp4mtbal	+0.006	-0.011	+0.006	+0.006	+0.002	+0.005

Reading Tables 3.1–3.3 and Figures 3.1–3.15.

- **Status is a usable detector.** On validation the best F1 is 0.80 (Table 3.2) at a threshold near 0.43, with precision 0.75 and recall 0.86; at that point `Exp4mt` makes 481 false positives and 240 false negatives out of 3,064 recordings (error 23.5 %). On TUAB test the validation threshold carries over well (Table 3.3): F1 0.82 at the validation $t^\star$ against 0.84–0.85 when $t^\star$ is chosen on test, with 111 of 125 abnormal recordings found, 34–36 false alarms and 14 misses (balanced error 17–18 %). Figure 3.2(test) shows the curves.
- **Clinical seizure and stroke are above the flag-everyone floor but far from a detector.** Clinical seizure reaches F1 0.25–0.27 against a floor of 0.12 (Figure 3.1(a)). Its best operating point flags about 400–500 recordings to find 80 of the 185 seizure recordings: about 4–5 false alarms per hit (Table 3.2). Stroke reaches 0.16 against 0.06. In Figure 3.3(b) precision stays below 0.4 at every recall.
- **A high ROC-AUC does not make a rare group detectable.** Electrographic seizure has ROC-AUC 0.96 (Table 3.1), yet its best F1 is 0.13–0.18, with an interval reaching 0.34. At prevalence 0.004, `Exp4mt`'s best point finds 5 of the 11 positives with a false-positive rate of only 1.3 %, but that is still 39 false alarms. Dementia (ROC-AUC 0.85–0.89) is similar.
- **Many groups sit at the floor.** For bipolar, schizophrenia, MS and "other", $t^\star$ falls to about 0.01: the "best" F1 is then reached by flagging most recordings (e.g. schizophrenia: 1,800 false positives for 9 positives), and it is barely above the flag-everyone floor (Figure 3.1(b)). These groups are not predicted.
- **The class-balanced loss changes the ranking, not the detection.** `Exp4mtbal` raises the macro ROC-AUC from 0.70 to 0.72 (Table 3.1), but the macro best F1 stays at 0.09 (Table 3.2). Its gains are on groups too rare to give a usable F1.

Reading Table 4.1: does the age prediction matter for pathology?

- **Removing the age prediction does not improve the diagnosis groups.** `Exp4cls` (EEG only, no age loss) reaches a macro ROC-AUC of 0.715 against 0.722 for `Exp4mtbal` (Table 3.1). Its macro PR-AUC (0.053 vs 0.045) and macro best F1 (0.095 vs 0.092, Table

3.2) are about equal. Each run selected its epoch on this very metric, yet none passes `Exp4mtbal`; the age head neither costs nor helps the groups.

- **Without age, the status gets slightly worse.** `Exp4cls` has validation ROC-AUC 0.813 [0.797, 0.829] against 0.820 [0.804, 0.836] (Table 3.1). On TUAB test at the validation threshold (Table 3.3) it finds 100 of 125 abnormal recordings (25 missed, 28 false alarms; F1 0.79), against 111 found with 14 missed for the age-predicting runs (F1 0.82). Its test ROC-AUC is 0.912 against 0.918–0.923. Predicting age seems to act as a mild regularizer for the status head.
- **Knowing the age helps the status most, but within the noise.** `Exp4clsage` is the best status model: on validation ROC-AUC 0.827 [0.812, 0.841] and best F1 0.80; on TUAB test ROC-AUC 0.931 and F1 0.83 at the validation threshold (111 found, 31 false alarms, balanced error 16.0 % against 17.0–19.4 %, Table 3.3). The validation intervals of all four runs overlap, so this is a direction, not a proven gain. It fits the clinical rule that an EEG is abnormal relative to the patient's age: slowing that is normal at 80 is abnormal at 30.
- **The age input does not help the diagnosis groups.** `Exp4clsage`'s macro ROC-AUC is 0.711. It is best on electrographic seizure (ROC-AUC 0.983, best F1 0.29 with interval up to 0.52, Figure 3.4), but with 11 positives. It loses on stroke (0.73 against 0.80 for `Exp4cls`) and TBI (0.69 against 0.76), where age is a strong prior that the model may over-use.
- **Single groups move by more than the runs differ overall.** `Exp4cls` reaches F1 0.26 on stroke and 0.20 on TBI against 0.16 / 0.10 for `Exp4mtbal` (Table 3.2, Figure 3.1). Their intervals are wide (stroke up to 0.35, TBI up to 0.32), so such per-group gains are not reproducible evidence until repeated with another seed.

Conclusion. The status is predicted well by every variant (validation ROC-AUC 0.81–0.83, TUAB test 0.91–0.93, about 80–89 % of abnormal test recordings found at the validation threshold). Neither dropping the age prediction nor giving the model the age changes the diagnosis groups beyond their noise (macro ROC-AUC 0.70–0.72). They stay limited by how rare their positives are: only clinical seizure, stroke and TBI clear the flag-everyone floor by a useful margin, and none is a usable detector on its own (best F1 $\leq$ 0.27).